

ECE238L Computer Logic Design

Lab 1: Comparator and 7-Segment Display

Section 003: Wednesday 2:00-4:45

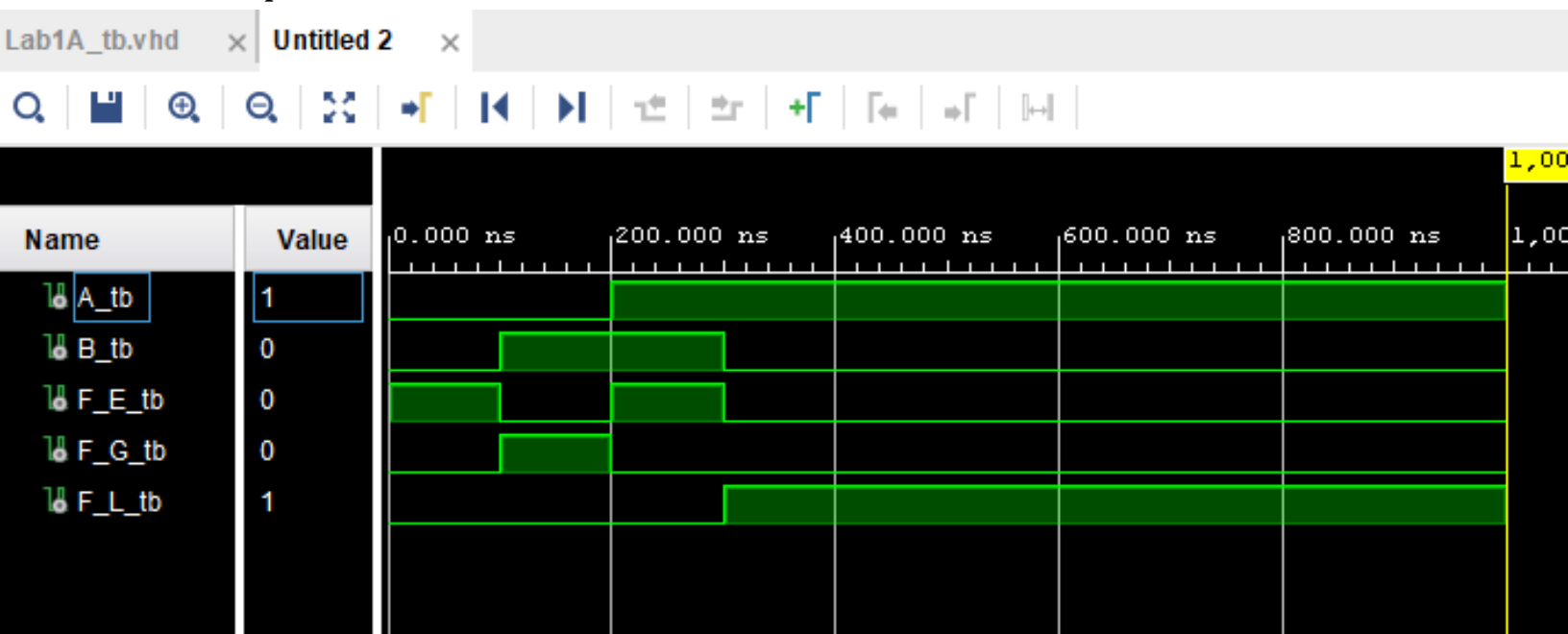
Name: Tomas Salaz

Lab 1 Comparator:

VHDL Source Code, Simulation Code (Test Bench) and Constraints

```
Lab1A.vhd
C:/Users/salaz25/ECE238/Lab1A/Lab1A.srscs/sources_1/new/Lab1A.vhd
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2 -- Company:
3 -- Engineer:
4
5 -- Create Date: 02/11/2025 01:12:50 PM
6 -- Design Name:
7 -- Module Name: Lab1A - Behavioral
8 -- Project Name:
9 -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12
13 -- Dependencies:
14
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
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21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity Lab1A is
35     Port ( A,B : in STD_LOGIC;
36           F_E : out STD_LOGIC;
37           F_G : out STD_LOGIC;
38           F_L : out STD_LOGIC);
39 end Lab1A;
40
41 architecture Behavioral of Lab1A is
42
43 begin
44     F_E <= '1' when B = A else '0';
45     F_G <= '1' when B > A else '0';
46     F_L <= '1' when B < A else '0';
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48 end Behavioral;
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Comparator Wave Form



Lab 1 7-Segment Display

VHDL Source Code, Simulation Code (Test Bench) and Constraints

Lab1B.vhd

C:/Users/tsalaz25/ECE238/Lab1B/Lab1B.srcs/sources_1/new/Lab1B.vhd

```
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity Lab1B is
35     Port ( A, B : in STD_LOGIC;
36           Cathode_7SD : out STD_LOGIC_VECTOR(7 downto 0);
37           Anode_7SD : out STD_LOGIC_VECTOR(7 downto 0));
38 end Lab1B;
39
40 architecture Behavioral of Lab1B is
41
42     signal F_E, F_L, F_G : STD_LOGIC;
43
44 begin
45     -- Logical expressions
46     F_E <= (not A and not B) or (A and B);
47     F_L <= A and not B;
48     F_G <= not A and not B;
49
50     process (F_E, F_G, F_L)
51     begin
52         if F_E = '1' then
53             Cathode_7SD <= "01100001";
54         elsif F_G = '1' then
55             Cathode_7SD <= "01000001";
56         elsif F_L = '1' then
57             Cathode_7SD <= "11100011";
58         else
59             Cathode_7SD <= "00000011";
60         end if;
61     end process;
62
63     Anode_7SD <= "11111110";
64
65 end Behavioral;
```

Lab1B_tb.vhd

C:/Users/tsalaz25/ECE238/Lab1B/Lab1B.srcs/sim_1/new/Lab1B_tb.vhd

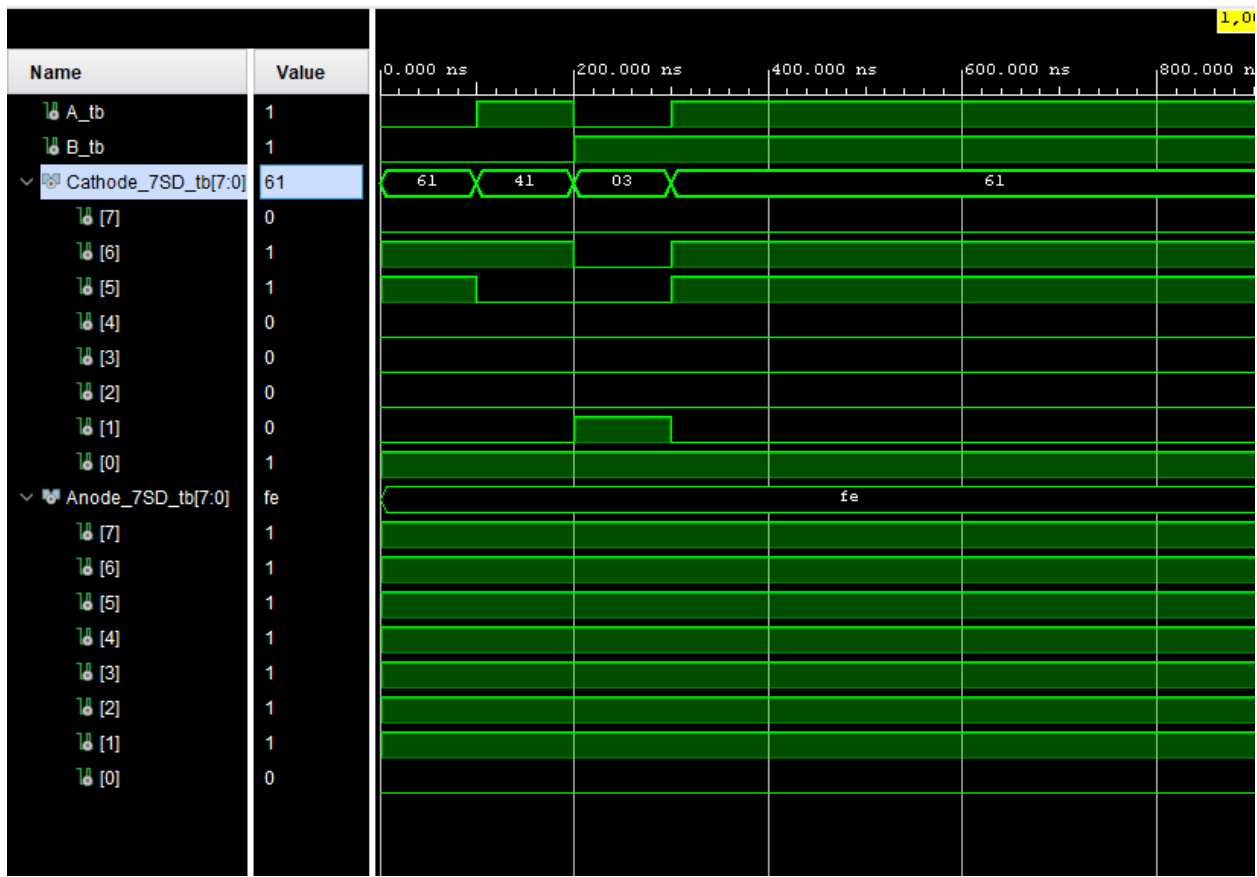
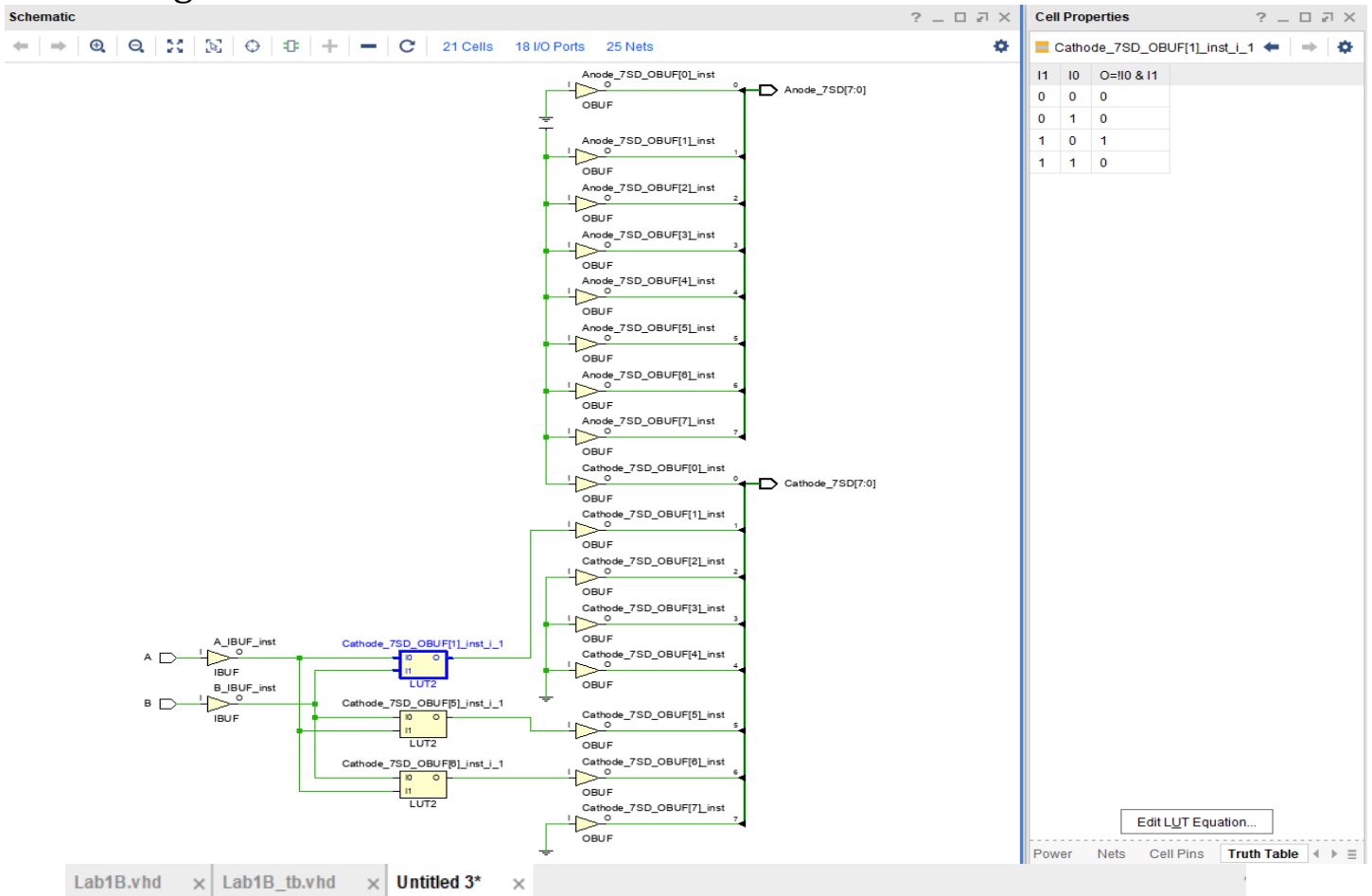
```
1 library IEEE;
2 use IEEE.STD_LOGIC_1164.ALL;
3
4 entity Lab1B_tb is
5     -- Port () ;
6 end Lab1B_tb;
7
8 architecture bench of Lab1B_tb is
9
10     component Lab1B is
11         Port (A, B : in STD_LOGIC;
12              Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
13              Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
14     end component;
15
16     signal A_tb, B_tb : STD_LOGIC;
17     signal Cathode_7SD_tb, Anode_7SD_tb : STD_LOGIC_VECTOR (7 downto 0);
18
19 begin
20
21     uut : Lab1B port map (A => A_tb,
22                          B => B_tb,
23                          Cathode_7SD => Cathode_7SD_tb,
24                          Anode_7SD => Anode_7SD_tb);
25
26     stimulus: process
27     begin
28         A_tb <= '0' ; B_tb <= '0' ; wait for 100 ns;
29         A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
30         A_tb <= '0' ; B_tb <= '1' ; wait for 100 ns;
31         A_tb <= '1' ; B_tb <= '1' ; wait for 100 ns;
32         wait;
33     end process;
34
35 end bench;
```

Lab1B_constr.xdc

C:/Users/tsalaz25/ECE238/Lab1B/Lab1B.srcs/constrs_1/new/Lab1B_constr.xdc

```
1 set_property PACKAGE_PIN J15 [get_ports A]
2 set_property PACKAGE_PIN L16 [get_ports B]
3
4 set_property IOSTANDARD LVCMOS33 [get_ports A]
5 set_property IOSTANDARD LVCMOS33 [get_ports B]
6 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[7]}]
7 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[6]}]
8 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[5]}]
9 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[4]}]
10 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[3]}]
11 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[2]}]
12 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[1]}]
13 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[0]}]
14 set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[7]}]
15 set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[6]}]
16 set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[5]}]
17 set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[4]}]
18 set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[3]}]
19 set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[2]}]
20 set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[1]}]
21 set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[0]}]
22
23 set_property PACKAGE_PIN J17 [get_ports {Anode_7SD[0]}]
24 set_property PACKAGE_PIN J18 [get_ports {Anode_7SD[1]}]
25 set_property PACKAGE_PIN T9 [get_ports {Anode_7SD[2]}]
26 set_property PACKAGE_PIN J14 [get_ports {Anode_7SD[3]}]
27 set_property PACKAGE_PIN P14 [get_ports {Anode_7SD[4]}]
28 set_property PACKAGE_PIN T14 [get_ports {Anode_7SD[5]}]
29 set_property PACKAGE_PIN K2 [get_ports {Anode_7SD[6]}]
30 set_property PACKAGE_PIN U13 [get_ports {Anode_7SD[7]}]
31
32 set_property PACKAGE_PIN H15 [get_ports {Cathode_7SD[0]}]
33 set_property PACKAGE_PIN L18 [get_ports {Cathode_7SD[1]}]
34 set_property PACKAGE_PIN T11 [get_ports {Cathode_7SD[2]}]
35 set_property PACKAGE_PIN P15 [get_ports {Cathode_7SD[3]}]
36 set_property PACKAGE_PIN K13 [get_ports {Cathode_7SD[4]}]
37 set_property PACKAGE_PIN K16 [get_ports {Cathode_7SD[5]}]
38 set_property PACKAGE_PIN R10 [get_ports {Cathode_7SD[6]}]
39 set_property PACKAGE_PIN T10 [get_ports {Cathode_7SD[7]}]
```

7-Segment Schematic and Truth Table and Waveform



Summary:

For Part A, we had to use VHDL to implement a Comparator using basic Logical Gates to compare if a bit was Equal, Greater Than or Less Than another bit. We only had 2 inputs so it was very simple and not complex at all. We also had to develop the code in VHDL so we could use the atrix machine and ensure that the functionality was implemented correctly.

For Part B, we used the Same logic as the Comparator to test only 2 bits against each other, but the output would be displayed using the 7-Segment display on the atrix machine. We use E,G and L to display the results, and we went through the same developmental process to ensure functionality.

Problems:

During the developmental process, tasks happened smoothly and the problems were very minor. VHDL syntax continues to be the main problem, but these problems are quick to sort out.

Hints:

Taking time to ensure the syntax is correct or using LLM's to find these syntax errors was helpful for me. I think asking the TA also helps since they have seen the language more and are more alert to the syntax.

Recommendations: None.